

WU, YU-HSUAN

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EDUCATION

National Yang Ming Chiao Tung University (NYCU)

2028 (expected)

Master of Arts in Linguistics

- Relevant Coursework (expected): Phonology, Computational Linguistics

National Taiwan University (NTU)

2026

Bachelor of Arts in Foreign Languages and Literatures

- Member of NTU Computational Linguistics Specialized Program
- Member of NTU Chinese-English Translation Program
- Relevant Coursework: NLP, Computational Linguistics, Phonology, Semantics

EXPERIENCE

Cathay Financial Holdings — DDT AI

Taipei, Taiwan

Feb 2026 – Present

ML/AI Engineer Intern

- Optimized AI News agent orchestration logic, data ETL, LINE integration and AWS deploy architecture.
- Built a recording-ASR-based BRD auto-revising agent with human-in-the-loop audit.
- Deployed a speech-driven form-filling service using streaming LLM on AWS for enterprise usage.

National Taiwan University

Taipei, Taiwan

Jun 2024 – Jun 2026

Research Assistant — Prof. Shu-hao Shih

- Designed perception experiment schema, prepared stimuli, and analyzed data for non-moraic schwa stress-bearing naturalness.
- Surveyed artificial-learning-experiment related phonology papers and prepared experiment stimuli using Praat and PsychoPy.

Taiwan AI Academy (AIA)

New Taipei City, Taiwan

Jun 2025 – Aug 2025

AI Developer & Technical Support Summer Intern

- Architected a LINE LIFF App career counseling chatbot with an n8n-based RPA workflow and managed user data with Supabase.
- Supported Enterprise Cloud Platform education courses as technical support.

PROJECTS

Voice-Notion — Cloud-First Standup Assistant (1)

Cathay Financial Holdings, DDT AI

- Built a Slack bot that converts mp4 oral work report into confirmable Notion pages and tasks, plus a daily automated standup summary
- Re-architected a single-instance Next.js service into a cloud-native ECS Fargate deployment with DynamoDB, SQS and EventBridge to enable horizontal scaling and zero-data-loss rolling updates
- Evaluated Lambda + API Gateway, self-managed EC2 and App Runner against ECS Fargate on ops cost and private-subnet criteria

AI News Agent — Three-Scenario Routing Chatbot (1)

Cathay Financial Holdings, DDT AI

- Built an internal AI research report query assistant using a four-layer architecture (Interface / Application / Data / External Services).
- Designed a three-scenario classification system (LLM intent detection + Python state machine) for intelligent routing across explicit, fuzzy, and out-of-scope queries.

- Built a full ETL pipeline (Google Sheets → Bedrock Embedding → OpenSearch) with multi-dimensional filtering by topic, source, date, and subscription tier.

BU Agent — Web-Based Two-Mode BRD Co-Drafting Assistant (1) Cathay Financial Holdings, DDT AI

- Built an internal web app using LangGraph + PostgresSaver with a split layout (chat ↔ mode-aware workspace)
- BU Agent serves as the BU's single entry point (two modes) to draft v1.0 BRD, then hands off structured deliverables to the downstream BA Agent.

PUBLICATIONS & TALKS

Examining LLM's Ability to Judge Presuppositions: A Comparative Study of Zero-Shot and Self-Discover-NLI Prompting ROCLING 2025

- Developed and presented the "Self-Discover-NLI" prompt engineering framework, adapting the three-step Self-Discover pipeline (SELECT → ADAPT → APPLY) specifically for Natural Language Inference (NLI) tasks.
- Conducted systematic experiments on the IMPPRES dataset (95 tasks per trigger type), evaluating GPT-4o's handling of clefts, conditionals, modals, and other linguistic triggers.
- Achieved ~10% overall accuracy improvement; performed in-depth error analysis identifying systematic model weaknesses.

SKILLS

Languages: Mandarin (Native), English (Fluent)

Programming: Python

Cloud & Infrastructure: AWS (Lambda, Bedrock, OpenSearch, DynamoDB, Amazon ECR, EventBridge, Amazon SQS)

Frameworks: LangGraph, LangChain

Tools: Microsoft Office, AWS CLI (register-task-definition, create-service, update-service, logs tail), Docker (node:20-slim image with ffmpeg, -no-cache builds)

Certificates: TOEIC (Listening and Reading 985, Speaking 180, Writing 170), AWS SAA (Solutions Architect Associate)